

Replication Plan

Constraining Cosmological Parameters with Needlet Internal Linear Combination Maps

I: Analytic Power Spectrum Formalism

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Auto-generated Replication Blueprint

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Paper Overview

This paper develops an analytic formalism for the power spectrum of Needlet Internal Linear Combination (NILC) maps used in cosmic microwave background (CMB) analysis. NILC is a component separation method that combines multi-frequency CMB observations to extract the CMB signal while minimizing foreground contamination. The paper derives analytic expressions for the residual foreground and noise biases in the NILC power spectrum, validates them against Monte Carlo simulations, and uses the formalism for cosmological parameter estimation (constraining parameters like $\Omega_b h^2$, $\Omega_c h^2$, H_0 , n_s , A_s , τ).

Detailed Workflow

Phase 1: Code and Data Setup

- 1.1 Clone NILC-PS-Model repository from GitHub.
- 1.2 Install dependencies: HEALPix/healpy, CAMB or CLASS, NaMaster, NumPy, SciPy.
- 1.3 Download or generate sky simulations.

Phase 2: Analytic NILC Power Spectrum Model

2.1 Implement needlet window functions.

- Define needlet bands $b_j(\ell)$ covering multipole range.
- Implement the cosine/Gaussian needlet profiles.

2.2 Compute ILC weights analytically.

- Multi-frequency covariance matrix per needlet band.
- Minimum-variance weights: $w_j = \frac{C_j^{-1} \mathbf{a}}{\mathbf{a}^T C_j^{-1} \mathbf{a}}$ where $\mathbf{a}$ is the frequency response vector.

2.3 Derive analytic NILC power spectrum.

- CMB signal contribution.
- Residual foreground bias term.
- Noise bias term.
- ILC bias from finite sky / needlet localization.

Phase 3: Monte Carlo Validation

3.1 Generate simulated multi-frequency sky maps.

- CMB (from CAMB/CLASS C_ℓ 's), synchrotron, dust, free-free, SZ.
- Instrument noise for each frequency channel (e.g., Planck-like or SO-like).

3.2 Apply NILC pipeline to each simulation.

3.3 Compute empirical NILC power spectrum and compare to analytic prediction.

3.4 Verify bias formulas against simulation average.

Phase 4: Cosmological Parameter Estimation

- 4.1 **Construct the likelihood** using the analytic NILC power spectrum model.
- 4.2 **Run MCMC** (emcee or CosmoMC) to sample the posterior of cosmological parameters.
- 4.3 **Reproduce parameter constraints** (contour plots, marginalized posteriors).

Datasets

Dataset / Source	Type	Location / Notes
NILC-PS-Model code + data	Repository	Check paper for GitHub URL
Planck sky maps	Public CMB data	https://pla.esac.esa.int/
Planck Sky Model (PSM)	Foreground simulations	https://pla.esac.esa.int/
CAMB/CLASS C_ℓ spectra	Generated	https://camb.info/ or https://lesgourg.github.io/class_public/class.html

Models, Tools, and Software

Tool / Library	Version	Purpose / URL
NILC-PS-Model	latest	Paper's analysis code GitHub (see paper)
HEALPix / healpy	≥ 1.15	Spherical harmonic transforms, sky map operations https://healpy.readthedocs.io/
CAMB	latest	CMB power spectrum computation https://camb.info/
CLASS	latest	Alternative Boltzmann solver https://lesgourg.github.io/class_public/class.html
NaMaster (pymaster)	latest	Pseudo- C_ℓ power spectrum estimation https://github.com/LSSTDESC/NaMaster
emcee	≥ 3.1	MCMC sampler
cobaya	latest	Alternative MCMC framework for cosmology https://cobaya.readthedocs.io/
NumPy / SciPy	≥ 1.21	Numerical operations
Matplotlib	≥ 3.4	Power spectrum and contour plots

Tool / Library	Version	Purpose / URL
GetDist	latest	MCMC chain analysis and triangle plots https://getdist.readthedocs.io/

Hardware Requirements

- Monte Carlo simulations: multi-core workstation (≥ 32 GB RAM for high-resolution HEALPix maps).
- MCMC: can run on a single machine but benefits from parallelization.